Health Informatics and Data Science Master's Degree Program at Georgetown University

Krithika Bhuvaneshwar[1], Adil Alaoui[1], Subha Madhavan[1], Peter McGarvey[1] and Yuriy Gusev[1]

[1]*Innovation Center for Biomedical Informatics, Georgetown University*

https://healthinformatics.georgetown.edu

PURPOSE

The demand for workers with health informatics skills is projected to grow at twice the rate of the overall employment in the US, but there is a serious shortage of qualified workforce in this field. To address these demands in workforce development, the **Innovation Center for Biomedical Informatics** in collaboration with **Biomedical Graduate Education (BGE)** at **Georgetown University** has launched a new Master's degree program in **Health Informatics and Data Science (HIDS).** The curriculum and syllabi are in compliance with Commission on Accreditation for Health Informatics and Information Management Education (CAHIIM) and the **American Medical Informatics Association (AMIA) core competencies.**

WHY HEALTH INFORMATICS ?

30% of world's stored data is generated in the healthcare industry (Health Catalyst, 2022)

Average annual pay for a Healthcare Data Scientist in the US is $114,514 (ZipRecruiter, Jan 2022)

Informatics market size 100 billion by 2030 with 12.2% growth rate

>50,000 Health informatics and Data Science open positions jobs posted on Indeed.com *(As of March 2022)*

THE PROGRAM

- Industry-driven, 12-month accelerated program
- Capstone project with industry partners
- 30 Credits full time or part time
- Real-world hands-on experience

Website: https://TinyURL.com/GUHIDS

THE GEORGETOWN DIFFERENCE

HANDS-ON EXPERIENCE
- Real world project experience in the classroom & through capstone projects with industry and government agencies

EXPERTS
- Courses taught by well-respected faculty who are practitioners and leaders in the field of health informatics

FLEXIBILITY
- 1-year accelerated, career-ready program with the flexibility to enroll part-time or full-time

IDEAL LOCATION
- Located in the nation's capital Washington D.C., which is home to numerous federal agencies (NIH, NCI, FDA, NIST) and 2,000 biotechnology companies

MULTI-DISCIPLINARY COURSEWORK
- Students interact with field experts, and faculty from top-ranked Georgetown University depts.

STUDENTS GAIN COMPETENCY IN

- Health data management
- Artificial intelligence & Machine learning applications
- Precision medicine analytics
- Utilizing data in Electronic Medical Records
- Medical imaging informatics
- Big data analytics

CAREER PATH EXAMPLES

- Clinical Informatician
- Data scientist or data analyst with deep domain knowledge in healthcare data
- Health data science team lead
- Data engineering expert in healthcare
- Bioinformatician
- Health Data Analyst
- Health business analyst/ consultant
- Nurse informatician
- Physician informatician
- Enter a PhD program

USING R AND BIOCONDUCTOR FOR TEACHING – EXAMPLE COURSES

Precision Health Informatics course

- Microarray Gene expression analysis
- RNA-seq gene expression analysis
- Visualization of results
- Genomic and bioinformatic data science analysis

Evidence based data analysis course

- Biostatistics analysis
- Graphical methods
- How to create reports using RMarkdown

Utilizing data in Electronic Medical Records

- Connect to an electronic medical record (EMR) using R
- Use SQL in Rstudio to generate summaries, output that into a tabular format in RMarkdown report
- Very efficient !

COURSE CURRICULUM

FALL
- Massive Health Data Fundamentals (HIDS-501)
- Utilizing Data in Electronic Medical Records (HIDS-502)
- Precision Health Informatics (HIDS-503)
- Evidence based data analysis in population health (HIDS-504)

SPRING
- Advanced Topics in Health Informatics (HIDS-505)
- AI for health applications (HIDS-506)
- Digital Health Applications (HIDS-507)
- Imaging Informatics (HIDS-509)

SUMMER
- Mandatory Capstone (HIDS-510)

MANDATORY CAPSTONE PROJECT

- In collaboration with industry partners and federal agencies agencies
- Summer Sessions I & II
- Apply what you learned through courses in the 2 semesters to develop a solution for a real world problem

INDUSTRY & GOVERNMENT AGENCIES ADVISORY PANEL

- Sage Bionetworks
- ESAC
- FDA
- Deloitte
- Flatiron Health
- COTA
- Syapse
- AstraZeneca
- TrinetX
- INOVA
- Meds
- Indivumed Group
- Argentys
- CVS Health
- PwC
- Invitae
- Open Source Electronic Health Record Alliance (OSEHRA)
- LexisNexis
- Assymetrik
- RadAmerica
- Cerner Corporation
- Medstar Institute for Innovation (MI2)
- Frederick National Labs

RECENT ALUMNI CURRENT JOBS

- *Informatics Specialist I at Mayo Clinic*
- *Data Programmer at Chemonics International*
- *Business Intelligence Developer at Deloitte*
- *Information Specialist at ERPi*
- *Research Data Analyst at the US Department of Agriculture (USDA)*
- *Senior Statistical Programmer at PRA Health Sciences*
- *Software Engineer at Essential Software (for Frederick National Labs)*

CONNECT WITH US !

WEBSITE: https://tinyURL.com/GUHIDS

https://tinyURL.com/HIDSyoutube

EMAIL US: hids@georgetown.edu

https://tinyURL.com/HIDSLinkedin

OUR ALUMNI: https://tinyurl.com/GUHIDSAlum

@Georgetown_HIDS